

AWI-ESM3 ↔ PISM orography and SMB coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-08-04

Abstract

Mission. Close the atmosphere half of the AWI-ESM3 ↔ PISM cycle in both directions: PISM's surface mass balance driven by the OIFS leg that just ran, and the ice-sheet surface elevation OIFS runs with following the ice sheet PISM produced. The ocean half was closed by the moving-cavity campaign; this is its companion.

Where we are. The SMB direction runs and the numbers hold up. A full coupled year produces a monthly surface mass balance file on the PISM grid, through OASIS, from the OIFS leg that just ran: $P - E - R = 147 \text{ mm w.e./yr}$ over the ice mask, against an observed Antarctic value of order 170. PISM does not consume the file yet, so the 130 ka taint of §1.1 still stands, and the orography direction has not started.

Timeline in §1, current state and open items in §2, dead claims in §3.2.

1 The campaign, in the order it happened

1.1 Taking stock: the atmosphere side is a stub (2026-08-03)

`couplings/oifs/just_copy_stuff.functions` copies `atmosphere_file_for_ice.nc` verbatim out of `a270122/lars_an`. Confirmed in `eval5`: a 192×96 ECHAM Gaussian grid carrying `temp2`, `tsurf`, `aprt`, `orog`, `swd`, `cc`, `q2m`, `tau`, `emiss`, `uv`, `TOAswd`.

TAINTED: every PISM leg of the moving-cavity campaign, `eval5` and `fgdbg02` included, ran dEBM on a 130 ka paleo ECHAM atmosphere. No PISM surface mass balance number from that campaign is a statement about AWI-ESM3. Ocean-side results are not affected; they are forced from FESOM output directly.

Two further findings from the same read. `suorog.F90:151-250` already implements both returns: `ECE_CPL_ISMM` reads `usurf_oifs` and patches `ZZOROG` through `REESPE/SPEREE`, `ECE_LANDICE` reads `plit_oifs`. That code is upstream (vendor `a2e46d5`, Jorge Bernales) and our `suorog.F90` is byte-identical to it. But we run `ECE_LANDICE` only, and the `plit` we write is not PISM's ice mask: `make_plit_oifs.py` derives it from the regenerated ICMGG land-sea mask capped at 60°S . `eval5/couple/antar_pism2ece.nc` holds one variable, `plit_oifs`, `nx=40320`. No `usurf_oifs`. So no PISM geometry reaches the atmosphere at all.

1.2 Choosing the ISM-mapper over XIOS (2026-08-03)

Everything dEBM needs is already in the XIOS output, and the de-accumulation pattern exists in `field_def_ljpg_safe.xml.j2`. Checked against `eval5` year 1903: `ssrd` = 6.845×10^5 and `tisr` = $1.226 \times 10^6 \text{ J m}^{-2}$ per one-hour step give 190 and 340 W m^{-2} ; `lsp+cp` = $1.13 \times 10^{-4} \text{ m/step}$ gives $3.14 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$. All correct global means, so a jinja-gated `file_def` block would have delivered the forward direction with no new component.

Decided against it. XIOS solves the forward direction only: it cannot deliver `plit` into the live atmosphere, and cannot route basal melt and discharge into the ocean. Both need a component inside

the OASIS exchange, and one mechanism is better than two. Staying on the EC-Earth component also keeps CISSEMBEL available.

Also decided: carry both SMB schemes with dEBM as default; SMB before orography; everything behind `general.smb_coupled`.

1.3 The ISM-mapper becomes an `esm_tools` component (2026-08-03)

`configs/components/ismm/ismm.yaml`, in the `develop-is` coupling, always built.

It is not a standalone repository: it lives inside `ecearth4` under `sources/ism-mapper`, so the component clones `ecearth4` and builds that subtree. The clone is 37 MB, not the 663 MB the local `.git` suggests, because every `sources/*` model is a submodule that is not recursed into. The branch `cissembel-levante-intel` is required rather than preferred: it carries 5799a2a, which fixes an `EBMfinalize` use-after-free that crashes any run with a non-CISSEMBEL region.

CISSEMBEL is a *link-time* dependency even on the direct path (`ism-mapper_run_mod.F90:7` has an unguarded `use modEBMworld`), so it is built from the same clone. At runtime `EBMworld` only fires for regions with `ISMrgncis(n)=.true.`, so it never executes for us.

The committed `Makefile` carries ECMWF HPC2020 paths. Everything it needs is a plain make variable and command-line macros beat file assignments, so the machine configuration is passed in `comp_command` rather than by patching a tracked file.

Verified. Built end to end from a fresh clone. The open risk was whether code written against OASIS 5.2 links against our 5.0-smhi; it does, with no source change. The binary carries `oasis_init_comp`, `oasis_def_var`, `oasis_enddef`, `ebmworld` and `ebminitialize`.

1.4 The PISM grid goes into `ocp-tool` (2026-08-03)

The ISM-mapper is an OASIS component, so the PISM domain must appear in `grids.nc/masks.nc/areas.nc`. This went into `ocp-tool`, not `esm_tools`: `ocp-tool` already sits between PISM and AWI-ESM3, it is where the weight generation lives, and `ocp-tool/grids/` already has the abstraction (a class with `cell_latitudes`, `cell_longitudes`, `cell_corners`, `cell_areas`, `cell_masks`). Putting it there deleted a planned step: `ocp-tool` writes `ismp` itself, so nothing appends fragments afterwards.

Centres come from the PISM file's own `lat/lon`; corners are inverse-projected from the cell corner (x, y) ; areas are spherical excess over two triangles.

Verified on `antarct_cr` 8 km, 761×761 , three ways, because a wrong grid fails as silently-bad weights rather than a crash:

- Area total 36.03 against 36.28×10^6 km² from the plane area corrected by the polar stereographic scale factor, -0.70% .
- Cell areas 67.0 km² at the pole to 53.8 at the outer corner against a nominal 64, the right sign and size for `lat_ts = -71`.
- All 579121 cells wind counterclockwise seen from outside the sphere, no sign flips. SCRIP requires this and the area check cannot detect it.

A full TCO95/CORE3 run puts `ismp` in all three files, bit-identical to an independently generated reference.

Found on the way: `ocp-tool` was appending the FESOM grid unconditionally. FESOM writes its own `feom` entry at runtime in domain-decomposition order; `ocp-tool`'s is mesh order, and the two disagree. Now behind `ocean.write_oasis_grid`, default off. *This changes default output for every FESOM config, not only ours.*

1.5 Declaring the OASIS coupling (2026-08-03)

Field names on the ice side come from the ism-mapper itself (cplng_config_mod.F90): <prefix>_Precip_liquid, _Precip_solid, _Evaporation, _Runoff, _Soil4T, _SST inbound and <prefix>_plit out. On the OIFS side A_Evap_ISM, A_SST_ISM and A_Soil4T_ISM are what ecearth.F90 declares for the direct path; precip and runoff are reused from the ocean path.

Two constraints the parser caught, both of which would have been expensive later:

- A_Runoff is on atmr, not atma, so it needs its own exchange line and its own direction. Grouping it with precip fails the same-grid validation.
- include_models is consumed before general-level choose_ blocks resolve, so ismm must sit in the develop-is add_include_models. It moves together with smb_coupled: a launched ismm with no coupled fields deadlocks OASIS.

1.6 Generating the weights (2026-08-03)

Weights are not left to OASIS at model runtime. ocp_tool/oasis_weights.py precomputes them with pyOASIS taken from the coupled model’s own OASIS build, ported from rdy2cpl: one MPI rank per link, each calling enddef() so SCRIP writes that link’s rmp_*.nc. Geometry is read back from the OASIS description files, so the ismp entry of §1.4 is exactly what a link consumes. It needs compute nodes; the one-rank-per-link parallelism is what makes it tractable.

Tested on 3 nodes against ocp-tool’s own TCO95/CORE3 output:

link	map	result
A096 → ismp	BILINEAR D	69 MB
R096 → ismp	GAUSWGT D	122 MB
ismp → A096	GAUSWGT LR	44 MB
ismp → A096	DISTWGT LR	34 MB
ismp → A096	BILINEAR	OASIS abort, rc=1
ismp → A096	CONSERV	OASIS abort, rc=1

Why the ice→atm conservative and bilinear maps die. 381 of the 579121 cells have a corner longitude spread above 180° along the polar seam, and the pole cell’s four corners sit at −135, 135, 45, −45, encircling the pole outright. That breaks SCRIP’s per-cell longitude line integral (CONSERV) and its enclosing-quadrilateral search (BILINEAR). Neighbour-based maps do not care. Same geometry as the reason the areas use spherical excess.

Consequence. plit reaches suorog as an interpolated value, not the area fraction ECE_LANDICE_THRESH is written to read. EC-Earth’s bilinear would give the same, so we are no worse than the reference, but the threshold semantics are looser than ideal.

1.7 Running it: ten legs to a plausible number (2026-08-04)

Ten test legs, ismtest1 to ismtest10. The first six were plumbing, the last four were physics. Plumbing, briefly: the ism-mapper binary was stale because it had been built by hand rather than through esm_master and did not declare the namelist flags the generated namelist.ismm set; ismp was missing from the chunk-1 staged grids because chunk 1 stages from pism_cavity_ini/oasis/ and not from the fixed-mesh pool, so both pools now carry it; and LSendToRunoffMapper gated the ADD_FLD call but not UpdateISMForcing, which then looked for a field that had never been declared.

The mask wipe. ismtest6 and ismtest7 died the same way, at the clamp in ece_updclic_cpl.F90. updtim runs every timestep, the ISM coupling period is 86400s and the OIFS timestep is 3600s,

so 23 steps out of 24 deliver nothing. On those steps the receive buffer still holds the -HUGE it was initialised to, and CPLNG2_EXCHANGE does not tell the caller that nothing arrived: it accepts OASIS_0k, which means *call succeeded, no transfer*, in the same branch as the five statuses that do mean data arrived. An OASIS restart does not save you. It is read correctly at $t = 0$ and overwritten on the next step.

We fixed it on our side with ECE_LANDICE_FROM_OASIS, default `.false.`, so the OASIS-driven mask update is opt-in and ECE_LANDICE keeps reading `plit_oifs` at leg init. That is a workaround, not a fix. Any other OASIS_IN field whose coupling period exceeds the timestep has the same exposure. Issue drafted for the SMHI tracker.

Temperatures ten times too large. `ismtest8` delivered `antar_Soil4T` around 2600 K. The `gauswgt_avg` method had been copied from `gauswgt_c` and inherited its CONSERV GLBPOS postprocessing, which rescales the target field so the global integral matches the source. Source and target cover different areas, so the ratio is the whole error: A096 active area $359.41 \times 10^6 \text{ km}^2$ over `ismp` active area $36.03 \times 10^6 \text{ km}^2$ is 9.975. CONSERV belongs on a flux you are closing a budget with, between grids that cover the same domain. Neither holds here. Removed.

Too wet and too warm: the source grid was ocean-masked. `ismtest9` ran clean and produced `antar_smb_1900.nc` with no sentinels, which is the first end-to-end success. The numbers were still wrong, and the way they were wrong took a second look to see, because a first pass mistook the precipitation units. `A_Precip_liquid` and `A_Precip_solid` are divided by `RH0H20` where they are set, so they are m/s. `A_Evap_ISM` is not, so it is $\text{kg m}^{-2} \text{ s}^{-1}$. Mixed units in one exchange line.

The comparison that matters is against OIFS's own gridded output from the same leg, over cells that are actually ice sheet, annual means in mm w.e./yr:

	P	E	R	T_{soil4}
OIFS <code>lsp+cp</code> , <code>e</code> , <code>ro</code> over its own ice cells	212	31	8.6	–
<code>ismtest9</code> , all ISM fields on <code>atma</code>	426	28.1	11.4	256.9 K
<code>ismtest10</code> , land fields on <code>atmr</code>	168	12.9	8.4	239.1 K

All three ISM-mapper rows are masked to PISM's own `thk > 10 m`, 215030 cells, $13.8 \times 10^6 \text{ km}^2$. `ismtest9` was 2.6 times too wet and 18 K too warm. Every ISM field had been declared on `atma`, the ocean-masked A096, where 95% of the cells south of 75°S are excluded; the GAUSWGT stencil then draws from what is left, which is the Southern Ocean. Ocean precipitation is heavier and the ocean is warmer, and both show up exactly that way.

`atmr` (R096) is the same 40320-point reduced Gaussian grid with the land mask instead, it is already declared in `oifs.grids`, `A_Runoff` already uses it, and the `R096` $\rightarrow$ `ismp` weights already exist. Moving `A_Evap_ISM` and `A_Soil4T_ISM` to it is a one-line config change. `A_Precip_liquid` and `A_Precip_solid` are not, because they are the same fields the ocean coupling uses, and the ocean needs them on `atma`.

FIELD@grid. EC-Earth solves this by naming a different grid per `namcouple` entry, which is what OASIS is built for. `esm_tools` cannot: `coupling_fields[field]["grid"]` holds one grid per field, full stop. The obvious workaround is to declare `A_Precip*_ISM` aliases in OIFS, but that means editing code AWI-ESM3 shares with EC-Earth in order to work around a weakness in our own runscript engine, which is the wrong direction.

So the syntax gained a per-link override. `A_Precip_liquid@atmr` in a `coupling_target_fields` line uses `atmr` for that line and leaves the field declaration alone. It is 30 lines in `coupler.py`: one splitter, applied at each of the three places that parse a coupling string. Verified against the generated `namcouple`, where the ISM line now reads `R096 ismp` and the ocean line still reads `A096 feom`. `esm_tools b0555452f`.

ismtest10: the first SMB we believe. $P = 168$, $E = 12.9$, $R = 8.4$, so $SMB = P - E - R = 147$ mm w.e./yr over the PISM ice mask, 99.4% of the precipitation solid, mean soil layer 4 at 239 K. Observed Antarctic surface mass balance is of order 170 mm w.e./yr and the area-mean surface temperature is around -35°C . The runoff agrees with OIFS's own ice-sheet `ro` to 3%, which is the sharpest single check we have that the remap is not distorting anything.

The one number sitting low is P : 168 against OIFS's 212 over its own ice cells. That gap is not necessarily an error. The two masks are different objects, PISM's `thk > 10` m at 8 km against an annual-mean snow depth threshold on the 192×400 remapped output, and the latter is coarser and coastal-biased where accumulation is heaviest. Worth closing properly before any mass-budget statement, but it is not a blocker for feeding PISM.

2 State of the coupling (2026-08-04)

2.1 Proven

- ISM-mapper builds from a clean clone and links against our OASIS 5.0-smhi.
- `ismp` grid written by `ocp-tool` into all three OASIS files, geometry verified three ways (§1.4).
- All three coupling links generate on compute (§1.6).
- The `esm_tools` config assembles: `ismm` in `process_ordering`, `rstism.nc` a recognised OASIS restart, four exchange lines resolving with the prefix substituted.
- A six-component OASIS `enddef` succeeds with ISMMAP alongside OIFS, FESOM, XIOS, LPJ-GUESS and the runoff mapper.
- A full year runs coupled and writes `antar_smb_1900.nc` on the PISM grid, monthly, with no sentinels and no NaNs (§1.7).
- Daily coupling with `LOCTRANS AVERAGE`, so 365 exchanges a year rather than 8760.
- The delivered SMB is physical: 147 mm w.e./yr over the ice mask, 99.4% solid precipitation, runoff within 3% of OIFS's own ice-sheet diagnostic (§1.7).

2.2 Open

- P is 168 against OIFS's 212 over ice-sheet cells. Probably the two masks, not a bug, but it has to be closed on a like-for-like footprint before any mass-budget claim.
- **Consume the SMB:** turn `{region}_smb_{year}.nc` into `atmosphere_file_for_ice.nc`, replacing the stub of §1.1. This is what retires the 130 ka taint.
- **HTESSEL glacier-tile refreezing** is unchecked. It decides whether the shelf-margin SMB is meaningful before PISM ever sees it.
- **A_Calving double-count.** OIFS caps snow at 10 m w.e. and sheds the excess to the ocean as calving. With PISM coupled the same snowfall goes two ways. Has to be settled together with the freshwater return.
- **orog for the downscaling reference.** `field_def.xml:176` has a `z` but it is in the `3D_m1` group, so model-level geopotential. Source it from the ICMGG with eccodes for param 129, as `make_plit_oifs.py` does for `lsm`.
- **usurf_oifs writer**, replacing the ICMGG-derived `plit` with PISM's own mask, and turning `ECE_ISM_OROG` on.

- **PLIT_antar is received on atma**, because `ismp` → R096 weights were never generated. Harmless while `ECE_LANDICE_FROM_OASIS` is off, wrong the moment it is turned on for the orography half.
- **Freshwater return**. PISM basal melt and discharge do not reach FESOM. Not needed for SMB or orography, but it is the remaining leg of the mass budget.
- **Field set is welded to scheme in ismm**. The direct path supplies 6 fields, dEBM needs 7 and shares only 2 of them. Splitting the two would unblock dEBM for Greenland and for warm scenarios where a positive-degree-day treatment stops being adequate.
- **atlas 0.31.1 → 0.39.0** came in with the vendor merge and is unbuilt.

3 Reference

3.1 Secondary fixes

what	where	commit
<code>mod_domain.F08</code> listed in <code>MOD_SRC0</code> but absent, CISSEMBEL unbuildable	ecearth4	41e0d8e
Runoff / runoff-mapper exchanges made optional (-huge trap)	ecearth4	f9d0349
<code>ECE_ISMM_SET_STATE</code> call had no <code>intfb</code> include	oifs-48r1	38c3ab0
<code>debm</code> missing from <code>add_other_components</code> , <code>esm_master</code> aborted	esm_tools	55a7935
<code>LSendToRunoffMapper</code> gated <code>ADD_FLD</code> but not <code>UpdateISMForcing</code>	ecearth4	94511fb
OASIS-driven land-ice mask update made opt-in	oifs-48r1	d23116d
<code>gauswgt_avg</code> inherited <code>CONSERV GLBPOS</code> from <code>gauswgt_c</code>	esm_tools	c47fea07e
Per-link source grid override, <code>FIELD@grid</code>	esm_tools	b0555452f

The `-huge` trap is worth stating once in full: receive buffers are initialised to `-huge`, `AccumulateECEForcingFields` adds every `OASIS_IN` field with no check that it is coupled, and an uncoupled field is never written. So a declared-but-unpartnered input accumulates `-huge` for a whole leg. This is the same failure class as the OIFS-side `-HUGE` first-send bug, arriving from the other end.

3.2 Falsified claims (kept, struck through)

- ~~Adding a PISM grid means teaching the per-leg regeneration to emit SCRIP weights to and from TCO95 on every cycle. This is the expensive part and the main schedule risk. Overstated: the weight machinery is general, so PISM costs a few Link entries.~~
- ~~OASIS computes the weights itself at runtime, and the PISM grid does not move, so there is no per-cycle work.~~ Wrong twice. Weights are precomputed via `pyOASIS` (§1.6), and the links do regenerate per mesh-change leg because the OIFS mask moves where PISM is.
- ~~`CONSERV` is the right map for `plit`, since `ECE_LANDICE_THRESH` reads an area fraction. It aborts on a pole-centred grid (§1.6). `GAUSWGT` is what works.~~
- ~~The ice → atm link is `BILINEAR_D`. The SCRIPR letter is the *source* grid type and `ismp` is logically rectangular, so `LR`; and `BILINEAR` fails here regardless.~~
- ~~`plit-active=0` in the log is the symptom of the mask being wiped. That line appears 363 times a leg in `eval5`, which runs fine. It is a normal diagnostic and says nothing about the crash. The evidence for the wipe is the clamp itself and the delivery arithmetic, not this line.~~
- ~~All ISM source fields belong on `atma`. `atma` is ocean-masked. Everything except SST belongs on `atmr` (§1.7).~~

3.3 Method notes

- `esm_runscripts -c` assembles the full runtime config and catches grid-consistency and component-loading errors. Both config bugs above were found this way, not by reading.
- `esm_master -check <target>` renders the build without executing it. It also revealed the pre-existing `debm` section-header failure.
- Weight generation must run from a shared filesystem. A first attempt with the working directory on `/tmp` died in 14s: `/tmp` is node-local on Levante, so the compute node could not write the job output.
- Grid winding is invisible to an area check, which takes an absolute value. Test it separately by dotting the corner-polygon normal with the outward radial vector.
- Do not guess the units of an OASIS field. Read the line that sets it. `A_Precip_*` are divided by `RHOH2O` and are m/s , `A_Evap_ISM` is not and is $\text{kg m}^{-2} \text{s}^{-1}$, and they travel in the same exchange line. Guessing wrong turned a factor of 2.6 into an apparent factor of 350 and pointed at the wrong bug.
- Compare a remapped field against the source model's own output over the *same* mask, not over a latitude band. South of 60°S the domain is mostly ocean, and precipitation over the Southern Ocean is twice the ice-sheet value, so a band mean hides the error the mask exposes.

3.4 Assets

what	where
Reference document	<code>esm_tools/ece4-pism-coupling.md</code>
PISM OASIS grid (reference copy)	<code>/work/ab0246/a270092/input/pism/oasis/</code>
Weight-generation test output	<code>/work/ab0246/a270092/tmp/ism_p_wgen{,2}/</code>
ocp-tool ice-enabled config	<code>configs/TC095_CORE3_ICE.yaml</code>

repository	branch	state
<code>esm_tools</code>	<code>awiesm3-is-orography-and-smb-coupling</code>	<code>c94d11f89</code> local
<code>ocp-tool</code>	<code>feat/pism-oasis-grid</code>	PR #57, draft
<code>oifs-48r1</code>	<code>movcav-landice+co2-concdriiven</code>	pushed
<code>ecearth4</code>	<code>cissembel-levante-intel</code>	pushed